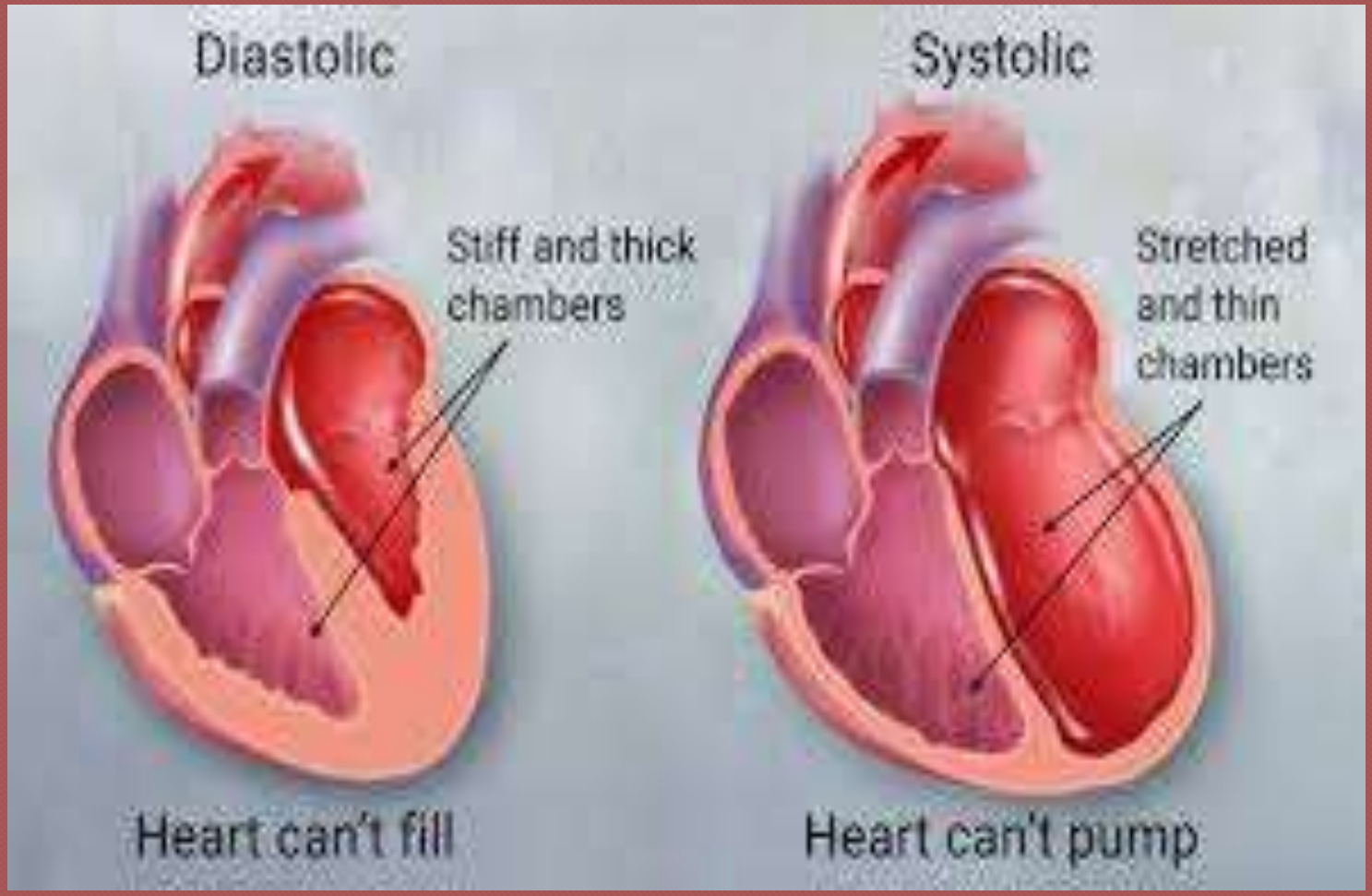
HEART FAILURE

APPLIED MEDICINE

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OVERVIEW OF HEART FAILURE

- Heart failure occurs when heart muscle doesn't pump blood as well as it should. Certain conditions, such as narrowed arteries in heart or high blood pressure, gradually leave heart too weak or stiff to fill and pump efficiently.
- Heart failure does not mean that heart has stopped or is about to stop working. It means that heart is not able to pump blood the way it should. It can affect one or both sides of the heart.


Diastolic
Stiff and thick chambers
Heart can't fill
Systolic
Stretched and thin chambers
Heart can't pump

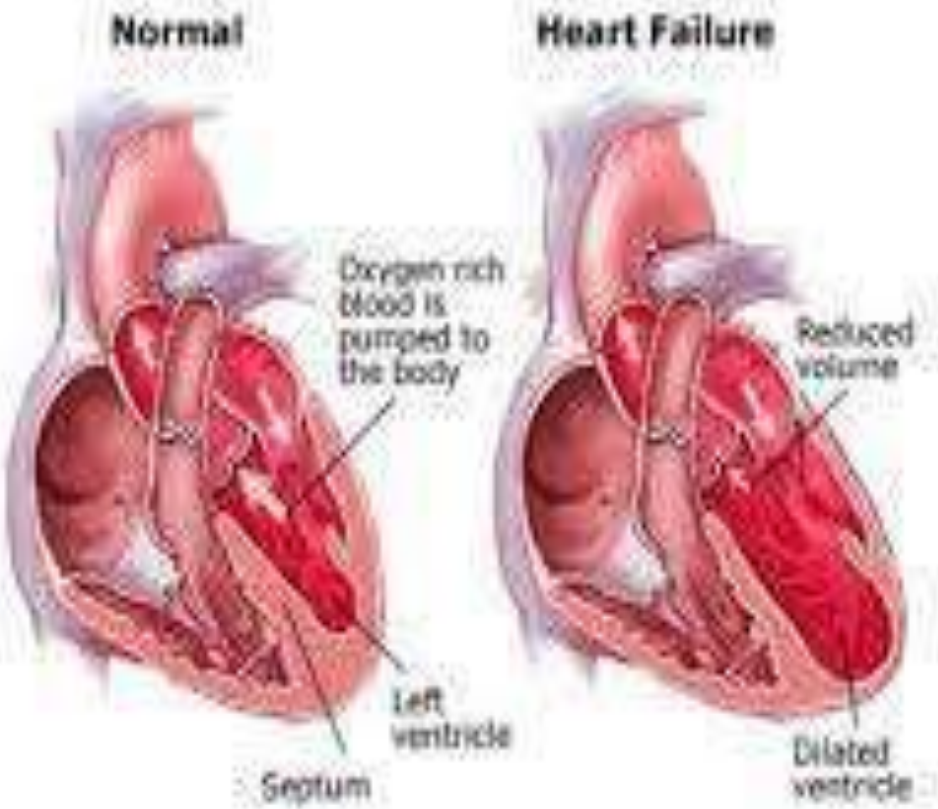
CAUSES OF HEART FAILURE

- Heart failure is caused by any conditions that damage the heart muscle. These include:
- **Coronary artery disease** - the coronary arteries supply the heart muscle with blood. If these are blocked or the flow is reduced, the heart does not receive the blood supply it needs.
- **Heart attack** - a sudden block of the coronary arteries; this causes scars in the heart's tissues and decreases how effectively it can pump.

TYPES OF HEART FAILURE:

- **Left-sided heart failure**
- The left side of the heart is responsible for pumping blood to the rest of the body. Blood backs up into the lungs as it is not effectively pumped away from the heart. This can cause shortness of breath and fluid buildup.
- **Right-sided heart failure**
- The right side of the heart pumps blood to the lungs where it collects oxygen. Right-side failure is, most often, caused by fluid build-up in the lungs due to left-side failure
- **Diastolic heart failure:**
- This occurs when the heart muscle is stiffer than normal. Because the heart is stiff, it does not fill up with blood properly; this is known as diastolic dysfunction.
- **Systolic heart failure:**
- Systolic dysfunction describes the heart's inability to pump efficiently after filling with blood. It often occurs if the heart is weak or enlarged.

Congestive Heart Failure

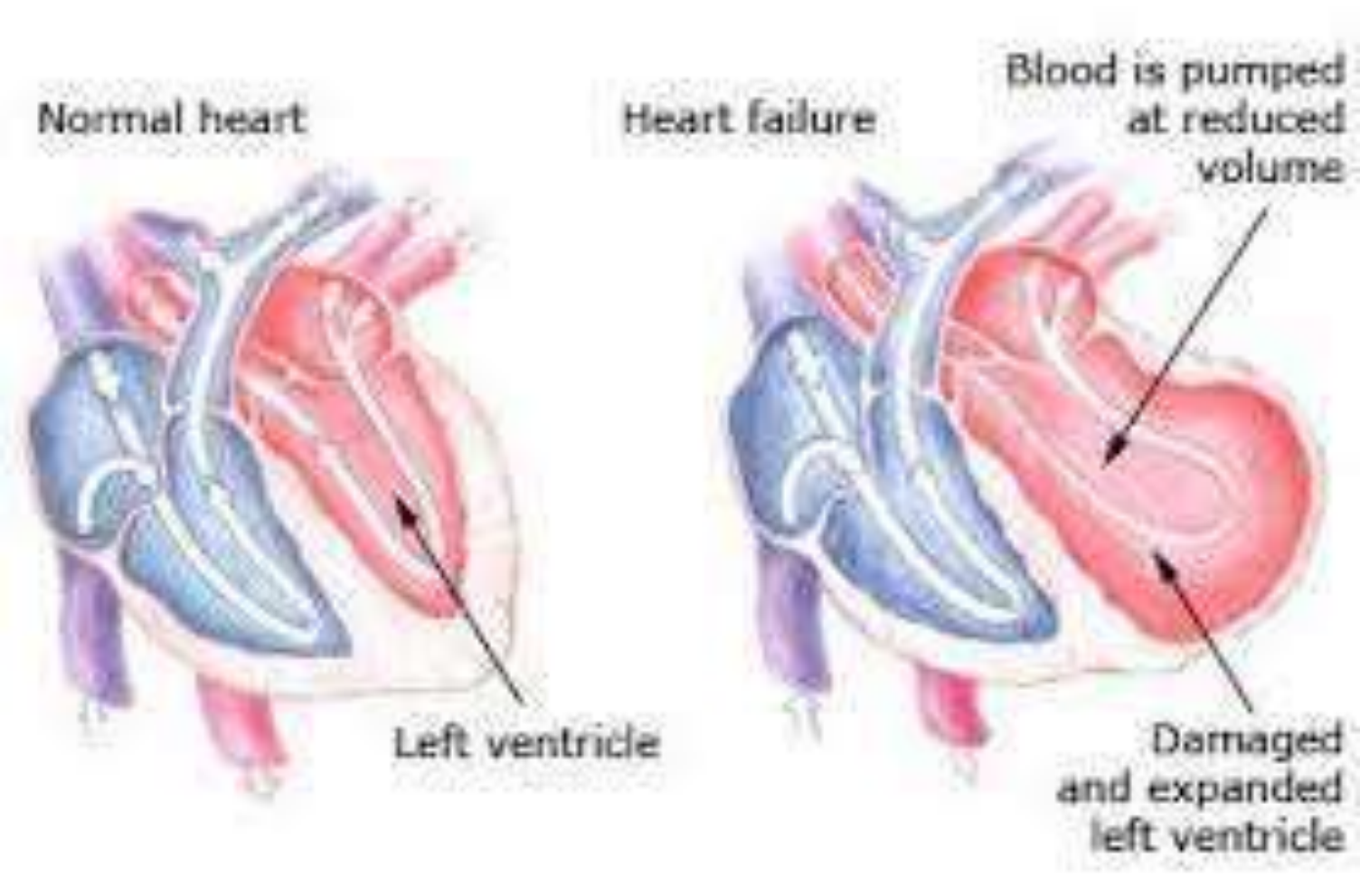


SIGN AND SYMPTOMS

- Shortness of breath (dyspnea
- Fatigue and weakness
- Swelling (edema) in legs, ankles and feet
- Rapid or irregular heartbeat
- Reduced ability to exercise
- Persistent cough or wheezing
- Increased need to urinate at night
- Swelling of abdomen (ascites)
- Very rapid weight gain from fluid retention
- Lack of appetite and nausea
- Difficulty concentrating or decreased alertness
- Chest pain if heart failure is caused by a heart attack

RISK FACTORS:

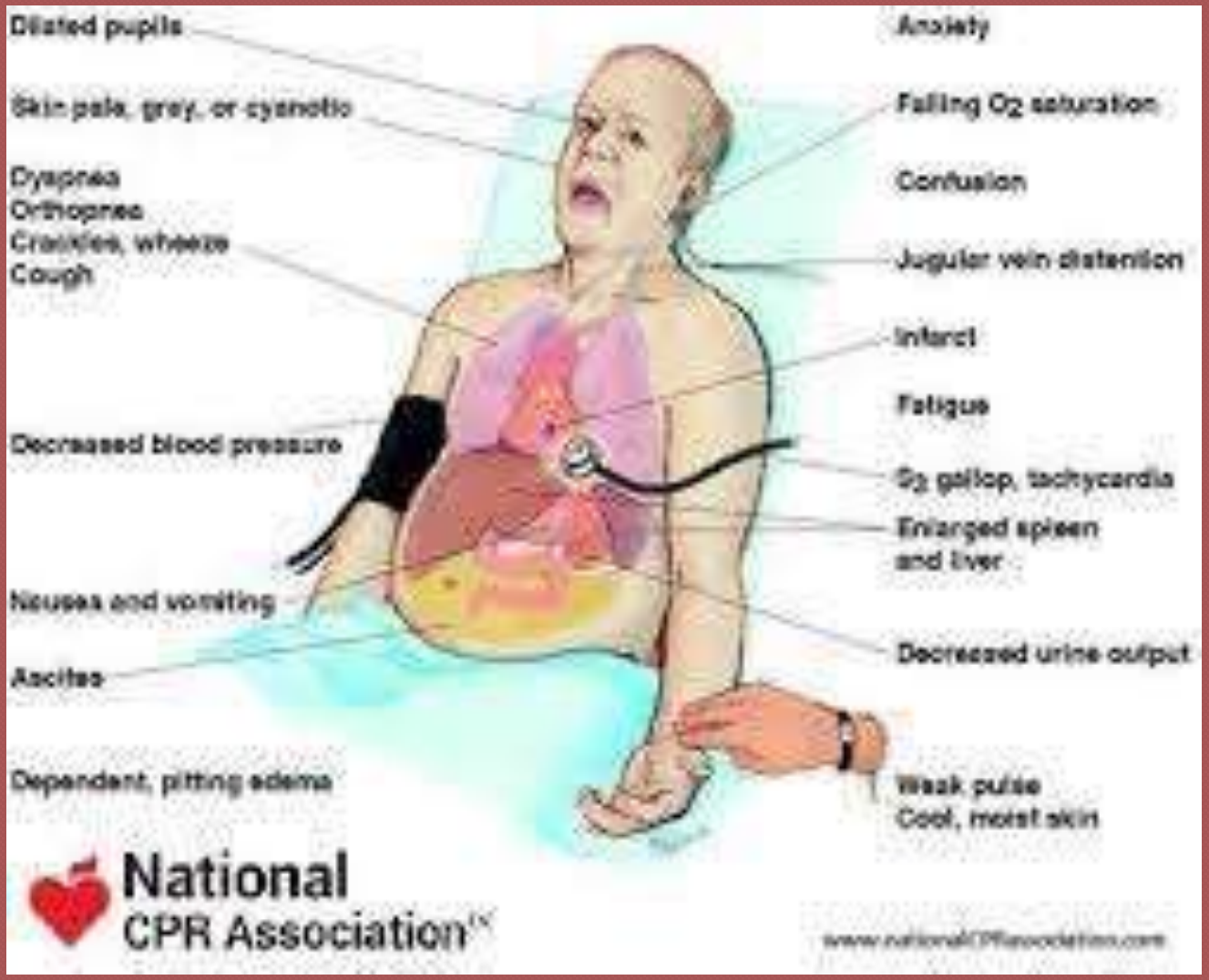
- High Blood Pressure
- Diabetes
- Irregular Heart beat(These abnormal rhythm, specially if they are very frequent and fast, can weaken the heart muscle and cause heart failure)
- Dyspnea (Shortness of breath)
- Use of Alcohol (Drinking too much alcohol can weaken heart muscle and lead to heart failure.)
- Smoking
- Irregular heart beat
- Viruses.


Normal heart
Left ventricle
Heart failure
Blood is pumped at reduced volume
Damaged and expanded left ventricle

TEST AND INVESTIGATIONS:

- **Electrocardiogram (ECG).** This test records the electrical activity of heart through electrodes attached to skin. It helps doctor diagnose heart rhythm problems and damage to heart

- **Stress test.** Stress tests measure the health of heart by how it responds to exertion. Patient may be asked to walk on a treadmill while attached to an ECG machine, or he may receive a drug intravenously that stimulates heart similar to exercise.
- Sometimes the stress test can be done while wearing a mask that measures the ability of heart and lungs to take in oxygen and breathe out carbon dioxide.


Dilated pupils
Skin pale, grey, or cyanotic
Dyspnea
Orthopnea
Crackles, wheeze
Cough
Decreased blood pressure
Nausea and vomiting
Ascites
Dependent, pitting edema
Anxiety
Falling O2 saturation
Confusion
Jugular vein distention
Infarct
Fatigue
S3 gallop, tachycardia
Enlarged spleen
and liver
Decreased urine output
Weak pulse
Cool, moist skin
National
CPR Association

TREATMENT (MEDICATIONS)

- **Angiotensin-converting enzyme (ACE) :** These drugs help people with systolic heart failure live longer and feel better. ACE inhibitors are a type of vasodilator, a drug that widens blood vessels to lower blood pressure, improve blood flow and decrease the workload on the heart. Examples include enalapril (Zestril) and captopril (Capoten).
- **Beta blockers.** This class of drugs not only slows heart rate and reduces blood pressure but also limits or reverses some of the damage to heart if patient suffer from systolic heart failure. Examples metoprolol (Lopressor).

SURGICAL TREATMENT:

- Coronary Bypass Surgery: In this procedure, blood vessels from leg, arm or chest bypass a blocked artery in heart to allow blood to flow through heart more freely.
- HEART VALVE REPAIR/REPLACEMENT:
- If a faulty heart valve causes heart failure, doctor may recommend repairing or replacing the valve. The surgeon can modify the original valve to eliminate backward blood flow.
- Valve replacement is done when valve repair isn't possible. In valve replacement surgery, the damaged valve is replaced by an artificial (prosthetic) valve.

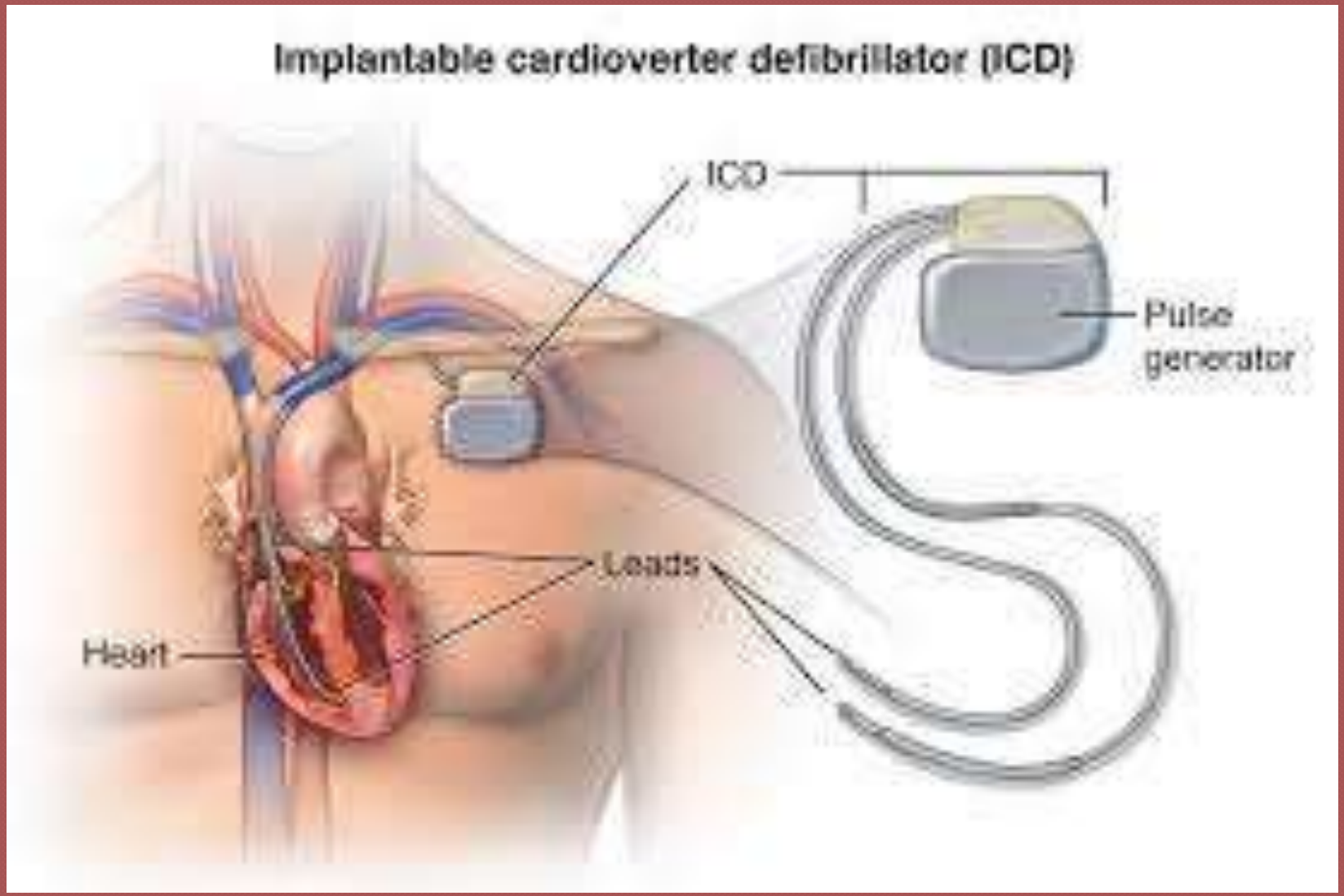
SURGICAL TREATMENT

- **Implantable cardioverter-defibrillators (ICDs).** An ICD is a device similar to a pacemaker. It's implanted under the skin in chest with wires leading through veins and into heart.

- The ICD monitors the heart rhythm. If the heart starts beating at a dangerous rhythm, or if heart stops, the ICD tries to pace heart or shock it back into normal rhythm. An ICD can also function as a pacemaker and speed heart up if it is going too slow.
- **Heart transplant.** Some people have such severe heart failure that surgery or medications don't help. They may need to have their diseased heart replaced with a healthy donor heart.
- Heart transplants can improve the survival and quality of life of some people with severe heart failure. However, candidates for transplantation often have to wait a long time before a suitable donor heart is found

COMPLICATIONS:

- **Kidney damage or failure.** Heart failure can reduce the blood flow to your kidneys, which can eventually cause kidney failure if left untreated. Kidney damage from heart failure can require dialysis for treatment.
- **Heart valve problems.** The valves of your heart, which keep blood flowing in the proper direction through your heart, may not function properly if your heart is enlarged or if the pressure in your heart is very high due to heart failure.
- **Heart rhythm problems.** Heart rhythm problems (arrhythmias) can be a potential complication of heart failure.
- **Liver damage.** Heart failure can lead to a buildup of fluid that puts too much pressure on the liver. This fluid backup can lead to scarring, which makes it more difficult for your liver to function properly.


Implantable cardioverter defibrillator (ICD)
ICD
Pulse generator
Leads
Heart

PREVENTION:

- Lifestyle changes can make to help prevent heart failure include:
- Not smoking
- Controlling certain conditions, such as high blood pressure and diabetes
- Staying physically active
- Eating healthy foods
- Maintaining a healthy weight
- Reducing and managing stress